

Assignment 8

Name: _____

Student ID: _____

13 problems. Show all of your work.

Problem 1.

Compute the partial derivatives of f with respect to x, y for

$$f(s, t) = s^2 - t^2, \quad s = \frac{x+y}{2}, \quad t = \frac{x-y}{2},$$

using both the chain rule and by plugging in s and t .

Problem 2. True or False

Consider $f(x, y) = (\cos x + (1+x)\tan y)e^{x^2-1+y}$. Then $\frac{d}{dt}f(t^2, t^3) = 0$ at $t = 0$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Problem 3. Homogeneous functions

A function f is called **homogeneous of degree n** if it satisfies the equation

$$f(tx, ty) = t^n f(x, y)$$

for all t , where n is a positive integer and f has continuous second order derivatives.

- Verify that $f(x, y) = x^2y + 2xy^2 + 5y^3$ is homogeneous of degree 3.
- Show that if f is homogeneous of degree n then

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf(x, y).$$

(Hint: use the Chain Rule to differentiate $f(tx, ty)$ with respect to t .)

- If f is homogeneous of degree n , show that $f_x(tx, ty) = t^{n-1}f_x(x, y)$ for $t > 0$.

Problem 4.

Let

$$w = 5xy + 2yz - 3xz, \quad x = st, \quad y = e^{st}, \quad z = t^2.$$

Compute $\frac{\partial w}{\partial s}(-4, -4)$ and $\frac{\partial w}{\partial t}(-4, -4)$.

Problem 5.

Consider $f(x, y) = e^{-r^4}$ where $r = \sqrt{x^2 + y^2}$. Compute its directional derivative at $(0, 0)$ with respect to the unit vectors in (polar) directions $\theta = 0, \frac{\pi}{4}, \frac{\pi}{2}$. What about any other angle?

Problem 6. True or False

If $\hat{u} = \langle a, b \rangle$ is a unit vector and f has continuous second partials, then

$$D_{\hat{u}}^2 f(x, y) = f_{xx}a^2 + 2f_{xy}ab + f_{yy}b^2,$$

where $D_{\hat{u}}^2 f = D_{\hat{u}}[D_{\hat{u}}f(x, y)]$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Problem 7. A mountain lion on a mountain

A mountain lion runs on a mountain whose height above the point (x, y) is $z = x^2 + \sin^2(xy)$.

- In which direction(s) should the mountain lion run from point $(1, 0, 1)$ so that the height is increasing at the fastest possible rate? What is this fastest rate?
- In which direction(s) should the mountain lion run from point $(1, 0, 1)$ so that the height is increasing at *half* of the fastest possible rate?

Problem 8.

Suppose that $f(x, y)$ is a function of two variables and let

$$\hat{u} = \frac{1}{\sqrt{2}} \langle 1, -1 \rangle, \quad \hat{v} = \langle 0, 1 \rangle.$$

Suppose that at a point $P_0(a, b)$,

$$\begin{aligned} (D_{\hat{u}}f)_{P_0} &= 0, \\ (D_{\hat{v}}f)_{P_0} &= 1. \end{aligned}$$

- Find *the direction* of the maximum rate of change at $P_0(a, b)$.
- Find the maximum rate of change at $P_0(a, b)$.

Problem 9.

Consider the function $f(x, y) = x^2 e^{3y}$ and the point $P_0 = (-1, 0)$.

- Find the direction in which $f(x, y)$ decreases the most rapidly at P_0 .
- Find the directional derivative of $f(x, y)$ at P_0 in the direction given by $\vec{v} = \langle -1, 1 \rangle$.

Problem 10.

Suppose the temperature (in degrees Celsius) at a point (x, y) is given by $T(x, y)$, where T is a continuous function that has continuous partial derivatives, and distances in the xy -plane are measured in meters. We are given the following information about T at the point $(x, y) = (1, 1)$:

- $T(1, 1) = 20$ °C,
 - the derivative of T at $(1, 1)$ obtains its smallest (i.e. most negative) value in the direction toward the point $(-1, 2)$,
 - the derivative of T at $(1, 1)$ in the direction of the vector $4\hat{i} + 3\hat{j}$ is 3 °C/m.
- A. Find the derivative of T at $(1, 1)$ in the direction toward the point $(4, -3)$.
- B. If you start at $(1, 1)$ and move along a straight path toward the point $(4, -3)$, *approximately* how far will you have to go in this direction until the temperature reaches 24 °C?

Problem 11.

f is a “nice” function of x and y (i.e. f is continuous and its partials exist). In Figure 1 are shown the level curves $f(x, y) = 0, 2, 4, 6, 8, 10, 12$.

- A. Let $\mathbf{u}_1 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$. Estimate $D_{\mathbf{u}_1}f(4, 3)$.
- B. Estimate $D_{\mathbf{u}_1}f(1, 2)$.
- C. Draw on the diagram a unit vector pointing in the same direction as $\nabla f(1, 2)$.
- D. At P , draw a unit vector and label it $\mathbf{u}_2$, such that $D_{\mathbf{u}_2}f(P) < 0$.
- E. Estimate $\frac{\partial f}{\partial x}(P)$.
- F. Which appears bigger, $\frac{\partial f}{\partial y}(Q)$ or $\frac{\partial f}{\partial y}(R)$?

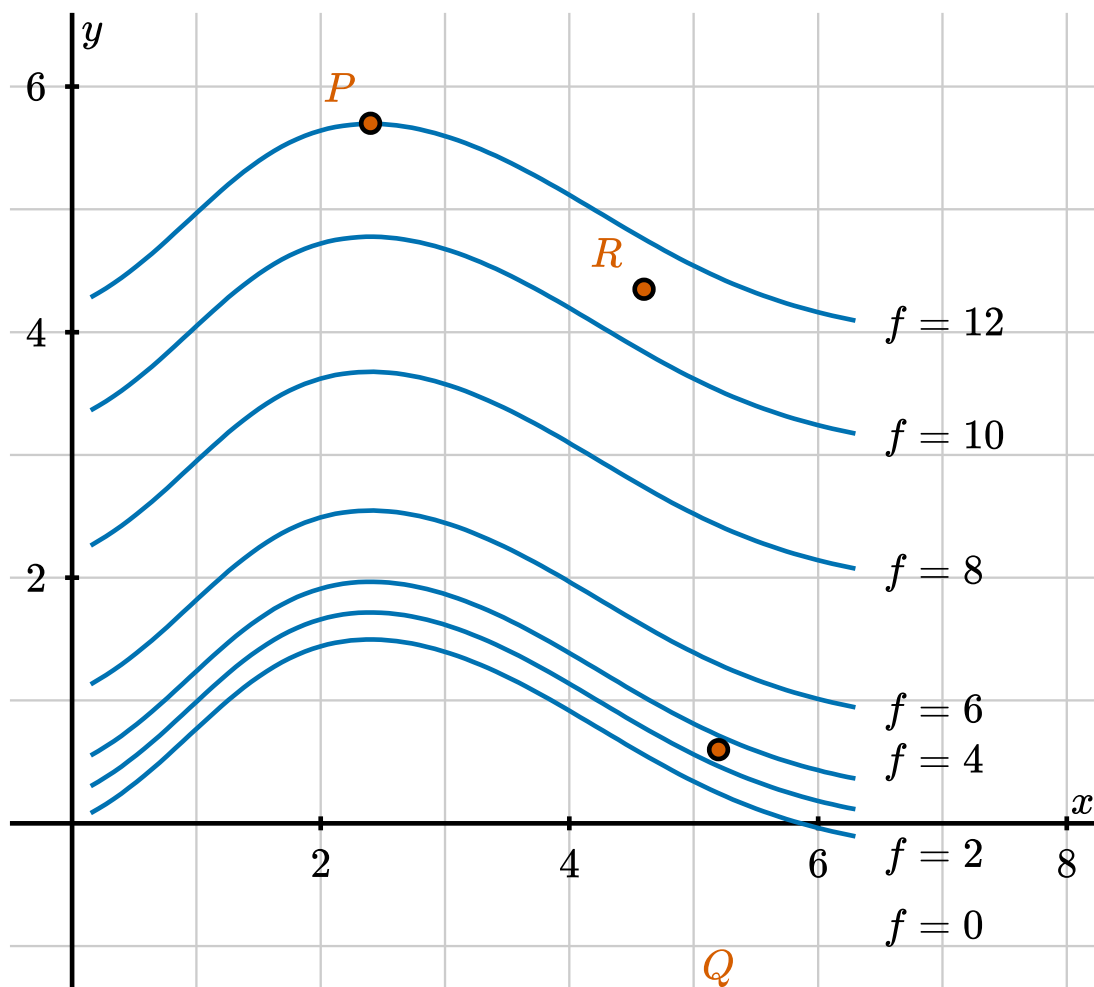


Figure 1. Level curves of f at the values $0, 2, 4, 6, 8, 10, 12$. The marked point P is the highest point of the $f = 12$ curve; Q lies midway between the $f = 2$ and $f = 4$ curves; R lies midway between the $f = 10$ and $f = 12$ curves.

Problem 12. *Electric Potential on a Coated Plate*

The electric potential on a thin coated plate, in volts, is

$$V(x, y) = 100 - 2x^2 - 6y^2,$$

with x and y in centimeters. Consider the point $P_0(2, 1)$.

1. Find ∇V at P_0 and the electric field $\mathbf{E} = -\nabla V$ there. Along which unit vector does the potential drop most rapidly, and at what rate?
2. A test charge is nudged along $\hat{\mathbf{u}} = \frac{4}{5}\hat{\mathbf{i}} - \frac{3}{5}\hat{\mathbf{j}}$. Compute $D_{\hat{\mathbf{u}}}V$ at P_0 .
3. Find the two unit vectors along which the potential does not change, and name the curve they are tangent to.

Problem 13. *Sailing the Isobars of a Low*

Near the center of a low-pressure system the sea-level pressure, in millibars, is modeled by

$$P(x, y) = 988 + 0.004(x^2 + y^2),$$

with x and y in miles from the center of the low. A ship sits at $P_0(20, 10)$.

1. Find the pressure at the ship and ∇P there. The pressure-gradient force per unit volume of air is a positive multiple of $-\nabla P$; in which unit direction does it push, and what is $\|\nabla P\|$ in millibars per mile?
2. A second ship 40 miles from the center reports a stronger wind. Using this model, explain why $\|\nabla P\|$ supports that report.
3. The captain decides to hold a constant barometer reading. Give the two possible unit headings at P_0 , and describe the shape of the full route.